
Feature Learning in Supervised AI Models

Zain Farhan

Lahore University of Management Sciences (LUMS)
Department of Computer Sciences

Abstract

In this research paper we will learn and investigate how various supervised learning models learn features. We will cover four path ways: Induction Biases, Domain Adaptation, Domain Generalization, and the Open Set Problem. The source code can be found on GitHub (<https://github.com/ZainFarhan-Work/ATML>).

1 Introduction

First, we will see the study the influence of Inductive Biases and Feature Representations, our goal here is to see the influence of model architecture. By comparing the metrics of four well known and standard models (ResNet-50, ViT-B, CLIP). We will first perform a baseline metric on all of them, and then perform 5 different experiments. Colour Bias, Grey Scale, Shape vs Texture, and Translation.

Then we will see Domain Adaptation and Domain Generalization, where we study DAN with MMD, DANN - Adversarial Alignment, CDAN, SAM, and DAN-DG. For the open set problem we will see GCSC and PROSER with both MLS and Placeholder

2 Experimental Setup/Methods

All experiments use seed 6304. All deviations from the manual are in Appendix A.

2.1 Inductive Biases and Feature Representations

- **Dataset:** STL-10.
- **Models:** ResNet-50, ViT-B/16 and CLIP ViT-B-32, frozen, with linear heads (plus zero-shot CLIP).
- **Style transfer:** For Shape vs. Texture we used the AdaIN implementation, which follows Huang and Belongie [2017]. The encoder/decoder architecture and the pretrained weights come from the public reimplementation [naoto0804/pytorch-AdaIN](#), with weights mirrored at [BhupatiNadar/adain-style-transfer-models](#) on HuggingFace.

2.2 Unsupervised Domain Adaptation

- **Data:** PACS. Source: Photo, Art Painting, Cartoon. Target: Sketch.
- **Model:** ResNet-18.
- **Optimisation:** AdamW, lr 10^{-4} , wd 10^{-4} .
- **Source-only:** Cross-entropy on the pooled source batch (also the ERM baseline for Task 3).
- **DAN** [Long et al., 2015]: $\mathcal{L} = \mathcal{L}_{cls} + \lambda \text{MMD}^2$ on 512-d features, kernel = sum of 3 RBFs with bandwidths $\{0.5, 1, 2\} \times$ median pairwise squared distance of the combined batch (bandwidth detached), $\lambda = 1$.
- **DANN** [Ganin et al., 2016]: gradient-reversal discriminator on the 512-d feature, $\alpha(p) = \frac{2}{1+e^{-10p}} - 1$.
- **CDAN** [Long et al., 2018]: as DANN, with discriminator input $\text{vec}(f \otimes p)$.

2.3 Domain Generalisation

- **ERM:** shared with Task 2. No Sketch access.
- **DAN-DG:** $\mathcal{L} = \mathcal{L}_{cls} + \frac{\lambda_{DG}}{3} \sum_{e < e'} \text{MMD}^2(F(X_e), F(X_{e'}))$, $\lambda_{DG} = 1$.
- **SAM** [Foret et al., 2021]: $\min_{\theta} \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}(\theta + \epsilon)$, $\rho = 0.05$.
- **Sharpness proxy:** $\Delta_{\text{sharp}} = \mathcal{L}(\theta + \epsilon) - \mathcal{L}(\theta)$, $\epsilon = 0.05 g / \|g\|_2$.

2.4 Open-Set Recognition

- **Data:** CIFAR-10 train 50,000 $\rightarrow$ stratified 90/10: 45,000 train / 5,000 val. Testing was done on both CIFAR-10 and CIFAR-100 (near and far unknowns, 800 each).
- **Model:** CIFAR ResNet-18: 3×3 stride-1 stem, no max-pool, random initialisation, 32×32 input, 512-d feature, Linear(512,10).
- **GCSC:** vanilla recipe plus RandAugment (2, 9).
- **PROSER** [Zhou et al., 2021]: initialised from Vanilla, 5 dummy classifiers, $\beta = 1$, $\gamma = 0.1$.
- **Scores:** MSP [Hendrycks and Gimpel, 2017] = $1 - \max_k p_k$. MLS [Vaze et al., 2022] = $-\max_k z_k$. Energy [Liu et al., 2020] = $-\log \sum_k e^{z_k}$. Mahalanobis = $\min_c (f - \mu_c)^\top \Sigma^{-1} (f - \mu_c)$.
- **Threshold:** 95th percentile of the score on CIFAR-10 validation.

3 Results

3.1 Inductive Biases and Feature Representations

We first take some basic measurements as a baseline, we freeze the three classifier heads and measure top-1 accuracy, macro-F1, and mean maximum confidence (Table 12). We also do this for zero shot CLIP.

Table 1: Overall Comprehensive Comparison

System	Clean			Grayscale			Hue +40°			Patch shuffle		
	Top-1	F1	Conf.	Top-1	Δ	Cons.	Top-1	Δ	Cons.	Top-1	Δ	Cons.
ResNet-50	96.4	96.39	0.91	94.6	-1.8	95.8	94.4	-2.0	95.2	89.0	-7.4	90.4
ViT-B/16	97.2	97.21	0.97	95.4	-1.8	98.0	95.8	-1.4	98.2	90.4	-6.8	92.6
CLIP + head	96.6	96.60	0.27	93.8	-2.8	93.0	94.8	-1.8	95.8	84.4	-12.2	84.6
CLIP zero-shot	97.0	97.01	0.94	93.0	-4.0	94.0	95.2	-1.8	95.6	84.4	-12.6	84.2

All the models perform well as a baseline, this maybe a limitation of the STL-10 dataset, as its distinct classes would be easier to classify.

3.1.1 Color Bias

We did two experiments with colour, the first we changed the images to greyscale measured the same metrics and got the following results (Table 13).

We can see that making the image greyscale results in a slight decrease in accuracy throughout. A more interesting observation is that there was no significant difference between Convolution and Vision Transformers, though consistency is slightly higher for the Vision Transformer. CLIP had the most sever impact. Both of the readings are lower than either ResNet or ViT-B, and further zero-shot is the lowest.

The second experiment was with a Hue Shift of 40 deg.

Again we see a minimal change in accuracy (Table 14), this time they are much closer to each other. No claims can be made as the difference is small enough that it can be considered noise. It is also of note that unlike greyscale, CLIP is much more in line with other models.

3.1.2 Shape vs. Texture

For this comparison alpha strength of 1 was required. We get the following results:

Out of a total of 400 candidate images only 286 were of good enough quality, which were further reduced to 220 to make each class balanced. As expected, the Vision Transformers have a considerable

Table 2: Shape vs Texture: Convolutional models are not shape biased.

System	Balanced ($n = 220$)					All accepted ($n = 286$)	
	N_{sh}	N_{tex}	N_{oth}	Shape bias	Coverage	Shape bias	Coverage
ResNet-50 + linear head	54	52	114	50.94	48.18	57.78	47.20
ViT-B/16 + linear head	99	14	107	87.61	51.36	89.94	55.59
CLIP ViT-B-32 + linear head	116	12	92	90.62	58.18	91.72	59.09
CLIP ViT-B-32 zero-shot	117	7	96	94.35	56.36	95.81	58.39

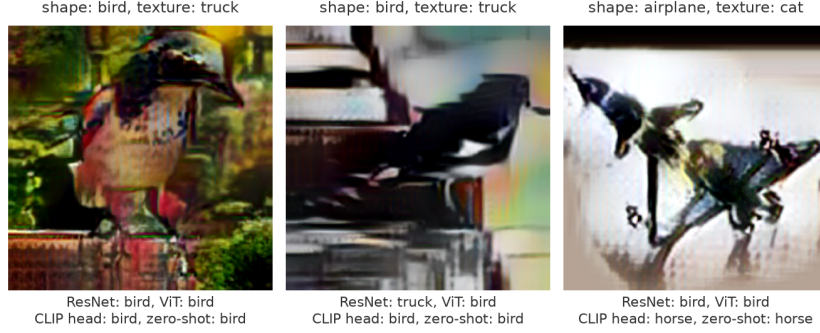


Figure 1: Example Images

shape bias, while the convolutional model does not. Though one point of note is that ResNet still managed to score a 50.94 on the Shape Bias. This is considerably higher than the original Geirhos et al. [2019] paper. The reason for this can be mostly likely presumed to be the biggest limitation of this experiment.

That is that the images were filtered out manually. This introduced significant human bias in the process making the data unreliable.

Another notable observation is that for ViT and ResNet both, there most quantitative category is neither shape nor texture, and CLIP barely has more shape classified images. Making the coverage very low, and limiting our ability to draw further conclusions from this data.

Together it is clear that colour reliance is very low, and cue conflict shows us that shape and texture reliance is high by comparison.. However the low coverage makes further analyses and stronger claims difficult.

3.1.3 Patch Structure + Translation

We divided the images into 4x4 patches, and shuffled the patches. We feed the shuffled images back in the model to measure the metrics (Table 15).

The ViT-B and ResNet model are close in accuracy.

One final thing of note is that CLIP performs noticeably worse than ViT, even though they share the visual encoder (as a Vision Transformer), the fact that both zero-shot and linear head land at the same number implies that its not the involvement of the text encoder in the prediction process. This suggest that contrastive learning (with the text encoder) during pre training or some other stage had some factor that made them more dependent factor.¹

The data for Translation can be seen in Figure3, from that it is clear that the spatial order is much more vital to the model than simply moving the image.

3.1.4 Representation Analysis

We measure the Cosine Similarity and t-SNE maps where relevant.

In terms of grayscale and translation we can see the Norm is rather high. This is consistent with our earlier hypothesis that models don't look at colour for classification.

¹We also saw more shape bias for them in table 2, but it was not enough of a difference to warrant any conclusion

Table 3: Cosine similarities - Expected Results

Intervention	ResNet-50		ViT-B/16		CLIP ViT-B-32	
	I_T	Norm.	I_T	Norm.	I_T	Norm.
Grayscale	0.789	0.719	0.737	0.711	0.900	0.743
Translation (32 px)	0.934	0.912	0.943	0.938	0.970	0.923
Patch shuffle	0.718	0.626	0.649	0.615	0.728	0.299
Cue conflict vs. content	0.250	0.004	0.223	0.146	0.651	0.101
Cue conflict vs. style	0.216	-0.041	0.072	-0.020	0.580	-0.083
Random-pair floor	0.247	0.000	0.090	0.000	0.612	0.000

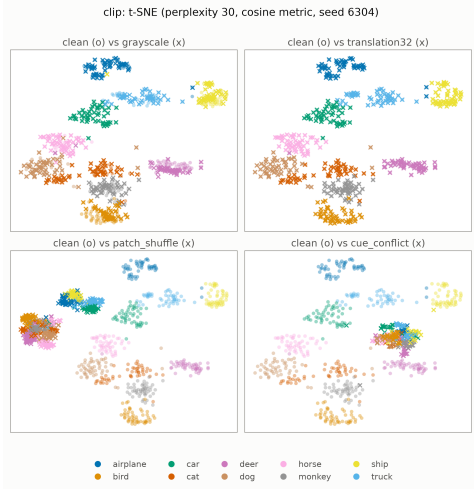


Figure 2: CLIP t-SNE: Representations over-lap except in Cue Conflict and Patch Shuffle

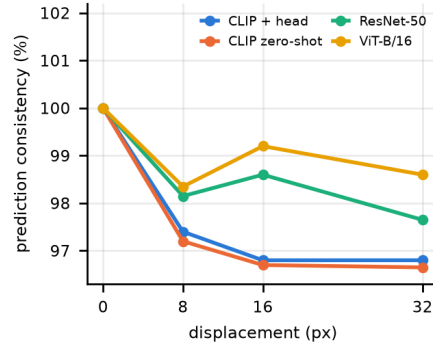


Figure 3: Translation.

In Patch shuffling the values a bit lower (around 0.6). What is more interesting is that CLIP and ViT show a large difference despite them both having a Vision Transformer. One plausible explanation is that since CLIP’s tokens are twice as large as ViT, they are more dependent on global structure, and the representation drifts more. This becomes more apparent in the following diagram.

The rest of the t-SNE diagrams (Figures 6, 7 in the appendix C.4) agree with these results.

Overall, a representation changes doesn’t always mean a prediction change. For grayscale they all have roughly the same Norm, but there is a noticeable difference between model performance, with ViT being the most stable. We’ve seen that though model architecture is not the full story as the two CLIP which are close still showed significant variance in results, it is still largely important, as ViT and CLIP were very shape biased, while CNN was not.

3.2 Unsupervised Domain Adaptation

3.2.1 Source Only ERM

We first train both the model and the classifier here on the source domains using cross entropy, which results in these measurements (Table 16).

As we can see in source only statistics, the accuracy on our source domains is very good, but when ran on a unknown domain (for us that is sketch) we notice a massive accuracy drop. compared with the mean it is a -33 % drop.

when we see the by class accuracy on the target domain (Table 17), dog performs worst while elephant performs best. Dog is the largest class in the sketch data, we found dog’s 333 most common errors go to elephant, 167 to giraffe. It is probable it is not confusing animals and objects at least, it is confused within animal classes.

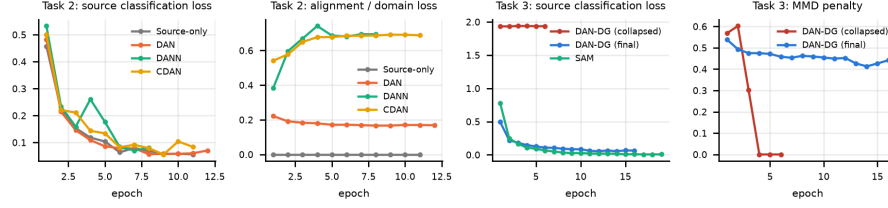


Figure 4: Training curves

Table 4: Diagnostic - Separability might not be the deciding factor

Method	Source val. acc. (%)			Mean source (%)		Sketch (%)		Δ Acc.	Sep. (%)
	Photo	Art	Cartoon	Acc.	F1	Acc.	F1		
Source-only	97.30	91.71	95.74	94.91	94.85	61.59	63.09	—	100.00
DAN	95.80	93.41	95.95	95.05	94.93	64.44	60.55	+2.85	81.87
DANN	95.80	92.93	93.60	94.11	94.03	55.97	58.23	-5.62	99.31
CDAN	97.30	91.46	95.52	94.76	94.66	74.65	73.85	+13.06	99.86

3.2.2 DAN - MMD Alignment

If we observe the general by class results for MMD, we see a slight but still meaningful improvement over the sketch accuracy (Table 18). Though the sketch F1 - macro decreases by a bit.

We see all classes improve in accuracy, except for giraffe and elephant (Table 19).

3.2.3 DANN - Adversarial Alignment

We now use adversarial learning to see how well it performs. And get the following results (Table 20).

Interestingly not only is the performance on the sketch pictures worse than source, but the predictions on source only is also worse. At first this might make it look like DANN didn't work, but after further investigation we find this (Table 21).

The Loss was indeed decreasing, and the Domain Accuracy was going to 0.5., what this means is that even if we fool the discriminator. that does not necessarily mean an increase in performance.

3.2.4 CDAN - Class-Conditional Adversarial Alignment

The performance on the source only metric stays the same, but the accuracy on sketch images received a significant improvement (Table 22). However, it was not lower domain separability that did this, as domain separability essentially remained the same.

3.2.5 Common Evaluation and Alignment Diagnostic

We would assume that bringing the domains closer and making them indistinguishable would be key. However aside from MMD which tries to directly to do this with the Loss function, a freshly trained logistical regressor can still separate the classes. And our best performance from CDAN barely shows any change, implying it might not be the most key factor.

Further data shows us that four classes show interesting behavior (Table 23). Dog (+20.99 to +43.14) and Person (+25.00 to +38.13) improve under every method, while Elephant and Giraffe degrade under every method. It almost seems like the model is trading the improvement in two for the others, no one improved all four.

Table 5: Per Class Comparison: Four classes share a trend

Class	Sketch accuracy (%)						Δ vs Source-only / ERM				
	Source-only	DAN	DANN	CDAN	DAN-DG	SAM	DAN	DANN	CDAN	DAN-DG	SAM
Dog	21.11	47.93	42.10	64.25	45.60	36.14	+26.82	+20.99	+43.14	+24.49	+15.03
Elephant	94.73	75.54	35.00	80.54	90.14	94.73	-19.19	-59.73	-14.19	-4.59	+0.00
Giraffe	77.95	61.62	51.53	62.95	49.27	71.85	-16.33	-26.42	-15.00	-28.68	-6.10
Guitar	70.07	74.18	98.36	89.31	92.11	96.71	+4.11	+28.29	+19.24	+22.04	+26.64
Horse	46.32	57.60	48.04	71.69	86.40	74.14	+11.28	+1.72	+25.37	+40.08	+27.82
House	83.75	100.00	100.00	100.00	92.50	96.25	+16.25	+16.25	+16.25	+8.75	+12.50
Person	61.25	86.25	98.12	99.38	50.00	75.62	+25.00	+36.87	+38.13	-11.25	+14.37

Table 6: Control study - too high a λ collapses the model

λ_{MMD}	Mean source F1 (%)	Sketch acc. (%)	Domain separability (%)
0 (source-only)	94.85	61.59	100.00
0.1	94.73	62.31	92.17
1	94.93	64.44	81.87
10	5.07	4.07	60.03

Table 7: Full Comparison - source

Method	Source val. acc. (%)			Mean source (%)		Worst (%)		Sketch (%)		Δ Acc.	Sep. (%)	Δ_{sharp}
	Photo	Art	Cartoon	Acc.	F1	Acc.	F1	Acc.	F1			
ERM	97.30	91.71	95.74	94.91	94.85	91.71	91.69	61.59	63.09	—	91.21	0.2095
DAN-DG	96.40	92.93	94.88	94.74	94.45	92.93	92.56	71.49	73.20	+9.90	70.05	0.2400
SAM	97.00	93.41	96.80	95.74	95.73	93.41	93.65	74.12	74.98	+12.53	89.01	0.1116

As alignment rises, the data converges and separability is reduced while sketch accuracy is improved a little. moderate alignment is helpful but too much destroys class information. we can use Macro-F1 without target labels as a signal.

3.2.6 Controlled Design Study

A higher λ means a higher penalty for difference between the source domain feature distribution. $\lambda = 10$ completely collapses the model. F1 Mean on source alone is enough to disqualify 10 as optimal even without consulting target domain labels.

We had Source-only reach 94.91% mean source accuracy against 61.59% on Sketch, a drop of 33.3 points. We saw that lower separability does not mean better prediction on target domain. And we saw class conditioning massively help in target domain accuracy.

3.3 Domain Generalization

We reuse the same ERM Baseline as last time, refereed to in table 16.

3.3.1 DAN-DG - Pairwise Source-Domain Alignment

In our initial configuration, which matched task with DAN for the sake of comparison, the model collapses (Table 24).

We speculate that this happens due to the model too aggressively trying to align the source domains (Table 26). So to remedy this we try to ramp up the penalty term (λ) from zero to one as the model trains instead of just starting from one (Table 25).

However this also collapsed, was visible from source-side metrics alone with source F1 5.07, feature std 0, MMD 0 by epoch 4. So as a last resort we increase the Bandwidth Gradient to 1 in the RBF Kernel, this finally results in a significant improvement to the sketch accuracy, however we can no longer compare these results to our earlier experiments.

DAN-DG did make source domains less distinguishable but that was not the main driver for improvement in sketch accuracy. The model completely collapsed with $\lambda = 1$.

3.3.2 SAM - Parameter-Space Stability

This worked much better and cleaner than our DAN-DG we see a obvious improvement on our target domain (Table 27).

3.3.3 Common Evaluation and Diagnosis

We compare all our numbers against each other, with one interesting trend that was also in Domain Adaptation, which that accuracies on source domains are not at all predictive of sketch accuracy, difference between source being minimal with SAM and ERM only, but sketch accuracy is massively improved by SAM.

Table 8: ρ Values - Collapses on 0.1

ρ	Source F1 (%)		Sketch (%)		Sep. (%)	Δ_{sharp}
	Mean	Worst	Acc.	Δ		
0 (ERM)	94.85	91.69	61.59	—	91.21	0.2095
0.01	94.59	91.48	63.58	+1.99	90.38	0.1702
0.05	95.73	93.65	74.12	+12.53	89.01	0.1116
0.1	12.75	10.68	4.07	-57.52	67.86	0.0091

Table 9: Unknowness Score Types - Mahalanobis performed worse while Energy was Best

Score	AUROC (%)			At the 95% validation threshold (%)			
	Near	Far	All	Test accept.	Near rej.	Far rej.	FPR near / far
MSP	80.98	89.66	85.32	94.34	28.25	45.38	71.75 / 54.62
MLS	79.42	89.84	84.63	94.77	30.62	54.62	69.38 / 45.38
Energy	79.43	89.92	84.67	94.84	31.38	54.88	68.62 / 45.12
Mahalanobis	79.00	91.22	85.11	94.62	27.25	50.12	72.75 / 49.88

We see that DAN does the most work for Source Domain Separability which is what MMD targets, while SAM does not really touch for domain separability but does reduce sharpness the most as we want it to. SAM performs very well compared to DAN.

We now have also seen that mean source and and worst source are not good indicators or predictors of performance on the target domain (sketch). SAM lowers the sharpness the most, but that only partly tells us how well the model will perform on sketch. It is also hard to tell whether target aware DAN or target free DAN is better. The best target-aware result (CDAN, 74.65) is essentially tied with the best target-free one (SAM, 74.12). DAN and DAN-DG have difference that makes the scenario less comparable. DAN-DG tries to bring sources together, while DAN brings source and domain together. The final DAN-DG also lets gradient flow through the kernel bandwidth, while DAN keeps it detached, so the kernels aren't identical.

3.3.4 Controlled Desgin Study

We varied the ρ values for SAM to find how far the trend extends

Sharp still increases as the shallowed are becomes a subset, but performance is completely destroyed which slowest that it can't be used as an accurate independent measure of performance for SAM.

Before this experiment we expected higher values of ρ to look more broadly and perhaps find shallower areas. and the sharpness value did fall, but the collapse alongside it was unexpected.

3.4 Open-Set Recognition (OSR)

We first train a very traditional ResNet classifier, with ten classes to get a Vanilla score. The results are this (Table 28).

We froze the Vanilla Model, and measured different unknowness scores on it (Table 29).

3.4.1 GCSC - Strong Closed-Set Classifier + MLS

For CIFAR 10 we see that both validation and test accuracy increase slightly (Table 30), but when we see the effect on unknown classifications, we see that they also improve across the board (Table 31).

3.4.2 PROSER - Classifier and Data Placeholders

On the CIFAR-10 data set we see similar levels of performance (Table 32), though we get there earlier, the same trend continues for Unkown classifications (Table 33).

3.4.3 Common Evaluation and Failure Analysis

All Unknowness score types had similar performance, in the Near rej stats, which would be the hardest, Mahalanobis performed the worst while Energy performed the best.

GCSC is the best performer, with significant improvement over Vanilla, both PROSER variants perform worse than Vanilla, the PROSER MLS can be chalked by to Non Gain from the Vanilla.

Table 10: Different Models - GCSC performs the best, both PROSER are worse than Vanilla

Model	Score	CSA	AUROC (%)			At the 95% threshold (%)		
			Near	Far	All	Test accept.	Near rej.	Far rej.
Vanilla	MLS	94.26	79.42	89.84	84.63	94.77	30.62	54.62
GCSC	MLS	95.38	81.84	92.13	86.99	94.35	37.62	64.00
PROSER	MLS	93.63	77.61	87.57	82.59	94.58	29.38	44.50
PROSER	Placeholder	93.63	74.36	84.82	79.59	94.66	27.25	41.37

Table 11: Failure Rates - Far are much easier than near

Group	Unknown class $\rightarrow$ predicted class	MLS score	τ	Accepted in class
Near	pickup_truck $\rightarrow$ automobile	-11.54	-5.72	91/100
Near	bus $\rightarrow$ truck	-12.36	-5.72	82/100
Near	wolf $\rightarrow$ dog	-11.37	-5.72	81/100
Far	wardrobe $\rightarrow$ cat	-11.53	-5.72	58/100
Far	mushroom $\rightarrow$ cat	-12.73	-5.72	55/100
Far	sunflower $\rightarrow$ bird	-10.99	-5.72	52/100

The -1.25 gap could be noise. but PROSER Placeholder shows amore apprent worse score. This is more apprent in Far rejections, where both perform considerably worse than Vanilla. We choose EPOCH 14 as that is the CIFAR 10 validation was highest, but at that early stage it is possible that the PROSER simply didn'T training enough to make significant progress, and the extra unknown class hampered the model's ability (as the logits would now be spread on one additional class), instead of helping it.

Detecting Far imges seems to be much easier throughout. In table 11 we can see that where as the worst Near are all wrongly classied at above 80%, the far are all less than 60%.

We saw it is much easier to accept near vs far, 555 of 800 near unknowns are accepted (69%), against 363 of 800 far (45%). Pickup truck, bus, and wolf got accepted most for near, while wardrobe, mushrooms, and sunflower for far. pickup $\rightarrow$ automobile, bus $\rightarrow$ truck, tractor $\rightarrow$ truck, wolf $\rightarrow$ dog were plausible failures while Wardrobe $\rightarrow$ cat, mushroom $\rightarrow$ cat and sunflower $\rightarrow$ bird were surprising. MSP is normalised confidence and discards logit magnitude. MLS and Energy keep magnitude. Mahalanobis uses feature-space distance, so it is the most independent signal. MLS and Energy were close to each other, and MSP were close to both, with Mahalanobis being furthest. We saw very good improvements from GCSC, while little difference from PROSER.

4 Conclusions

We saw that in terms of inductive bias, CNNs have strong texture bias while Visual Transformers have a shape bias. Neither had strong colour or translation Bias. For Domain Adaptation we see DAN with MMD beat DANN with adversarial alignment while CDAN did better than both. With DOMain Adaptation we saw SAM perform much better than DAN-DG as well as being much more stable. Lastly, with our Open Set problem we saw GCSC significantly out perform, while both PROSER variants were unable to beat even vanilla

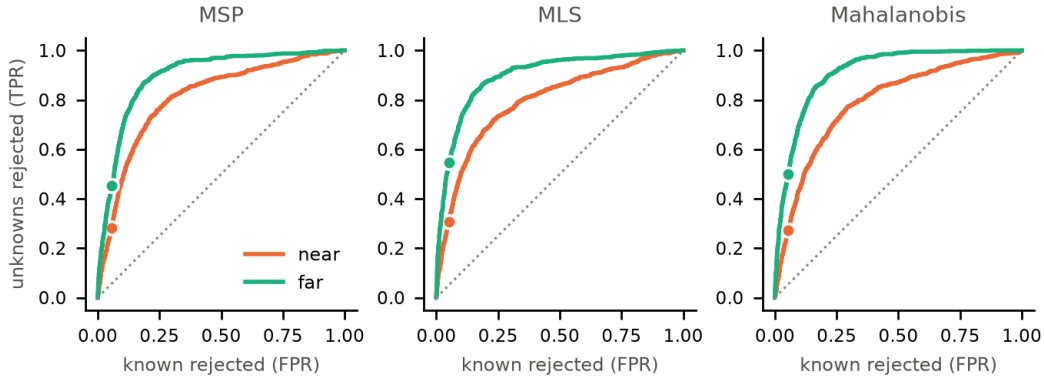


Figure 5: ROC curve for Detector - Far always performs better

References

- Pierre Foret, Ariel Kleiner, Hossein Mobahi, and Behnam Neyshabur. Sharpness-aware minimization for efficiently improving generalization. In *International Conference on Learning Representations (ICLR)*, 2021.
- Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016.
- Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A. Wichmann, and Wieland Brendel. ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. In *International Conference on Learning Representations (ICLR)*, 2019.
- Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations (ICLR)*, 2017.
- Xun Huang and Serge Belongie. Arbitrary style transfer in real-time with adaptive instance normalization. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.
- Weitang Liu, Xiaoyu Wang, John D. Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Mingsheng Long, Yue Cao, Jianmin Wang, and Michael I. Jordan. Learning transferable features with deep adaptation networks. In *International Conference on Machine Learning (ICML)*, 2015.
- Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I. Jordan. Conditional adversarial domain adaptation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- Sagar Vaze, Kai Han, Andrea Vedaldi, and Andrew Zisserman. Open-set recognition: A good closed-set classifier is all you need? In *International Conference on Learning Representations (ICLR)*, 2022.
- Da-Wei Zhou, Han-Jia Ye, and De-Chuan Zhan. Learning placeholders for open-set recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

Appendix

A Deviations and Caveats

- **Mirrors.** STL-10 verified byte-identical; PACS and CIFAR mirrors could only be checked by counts (class and domain totals).
- **Task 2 discriminator lr** 10^{-3} vs the specified 10^{-4} (adversarial methods only).
- **Task 3 DAN-DG headline** uses the bandwidth-gradient MMD, which differs from Task 2’s DAN implementation; the as-specified collapse is documented. A like-for-like comparison would re-run Task 2’s DAN with the same flag.
- **Task 3 separability probe** is class-weighted but not count-balanced across the three domains (manual: “balanced features”).
- **Task 4** augmentations run on uint8 tensors, not PIL; crop and flip are identical, RandAugment resampling can differ in rounding. PROSER losses were checked against the paper’s Eq. 4, 5 and 7, not against the authors’ code.

B Table Notes

Table 1. Top-1, macro-F1 (F1), Δ Top-1 and consistency (Cons.) in %; Conf. is mean max confidence. Δ and Cons. are relative to each system’s own clean predictions. CLIP is OpenCLIP ViT-B-32; the other systems use linear heads on frozen features.

Table 4. Sep. = domain separability: held-out accuracy of a logistic-regression probe ($C = 1$, balanced, 70/30 split, seed 6304) trained to tell source from target from the frozen features. Chance is 50%. Target labels were used only at this stage.

Table 9. Test accept. is the share of CIFAR-10 test images accepted at the threshold.

Table 10. CSA is closed-set accuracy on the ten known-class logits, so both PROSER rows share it. Test accept. is the share of CIFAR-10 test images accepted at the threshold.

Table 11. Vanilla, MLS; accept when the score is $\leq \tau$. Each row is the most confidently accepted image of that unknown class; the last column gives how many of its 100 test images were accepted.

Table 7. ERM is Task 2’s source-only checkpoint, reused unchanged. SAM uses the specified $\rho = 0.05$. DAN-DG uses $\lambda_{DG} = 1$ with gradient flowing through the median-heuristic bandwidth; with the bandwidth detached the run collapsed (Table 26).

Table 7. Separability: three-way logistic-regression probe over Photo / Art Painting / Cartoon on frozen features, $C = 1$, 70/30 split, seed 6304; chance is 33.3%. Δ_{sharp} : cross-entropy increase after one normalised ascent step of radius 0.05 on a fixed batch of 32 images per source domain, model in eval mode.

C Task 1: Additional Results

C.1 Clean Baseline

Table 12: Clean Baseline - Shows all models performing very well on the dataset.

System	Top-1 (%)	Macro-F1 (%)	Mean max conf.
ResNet-50 + linear head	96.4	96.39	0.91
ViT-B/16 + linear head	97.2	97.21	0.97
CLIP ViT-B-32 + linear head	96.6	96.60	0.27
CLIP ViT-B-32 zero-shot	97.0	97.01	0.94

C.2 Color Bias

Table 13: Greyscale: Minimal change in Top-1 Accuracy

System	Top-1	Macro-F1	Δ Top-1	Consistency	Conf.
ResNet-50 + linear head	94.6	94.61	−1.8	95.8	0.91
ViT-B/16 + linear head	95.4	95.41	−1.8	98.0	0.96
CLIP ViT-B-32 + linear head	93.8	93.78	−2.8	93.0	0.25
CLIP ViT-B-32 zero-shot	93.0	92.97	−4.0	94.0	0.93

Table 14: Hue Rotation of 40: Minimal change in accuracy

System	Top-1	Macro-F1	Δ Top-1	Consistency	Conf.
ResNet-50 + linear head	94.4	94.42	−2.0	95.2	0.90
ViT-B/16 + linear head	95.8	95.79	−1.4	98.2	0.96
CLIP ViT-B-32 + linear head	94.8	94.78	−1.8	95.8	0.26
CLIP ViT-B-32 zero-shot	95.2	95.18	−1.8	95.6	0.93

C.3 Patch Structure

Table 15: Patch Shuffling: Transformers suffer the most

System	Top-1 (%)	Macro-F1 (%)	Δ Top-1	Consistency (%)	Mean max conf.
ResNet-50 + linear head	89.0	89.06	−7.4	90.4	0.87
ViT-B/16 + linear head	90.4	90.52	−6.8	92.6	0.89
CLIP ViT-B-32 + linear head	84.4	84.18	−12.2	84.6	0.20
CLIP ViT-B-32 zero-shot	84.4	84.20	−12.6	84.2	0.82

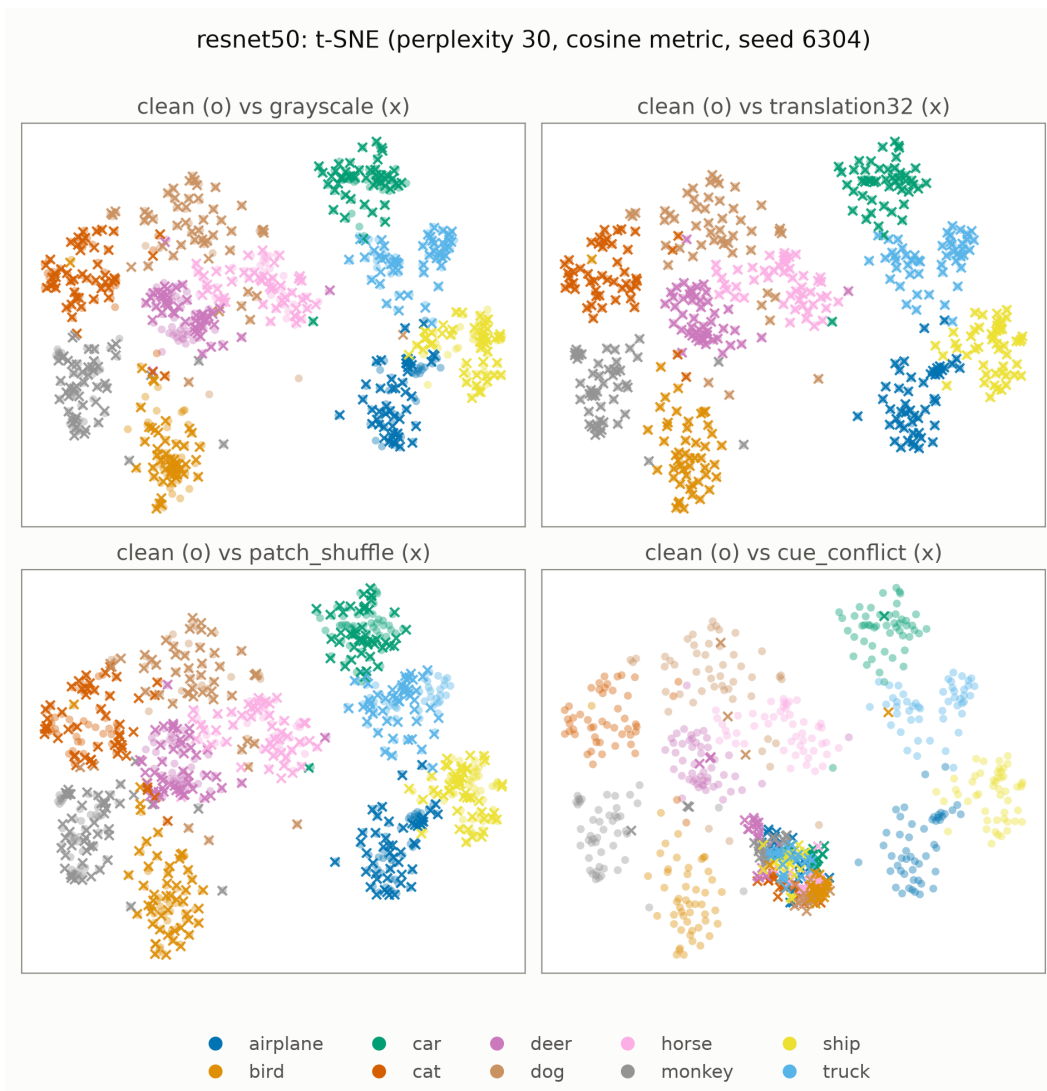


Figure 6: ResNet t-SNE: Representations overlap except in Cue Conflict

C.4 Representation Analysis

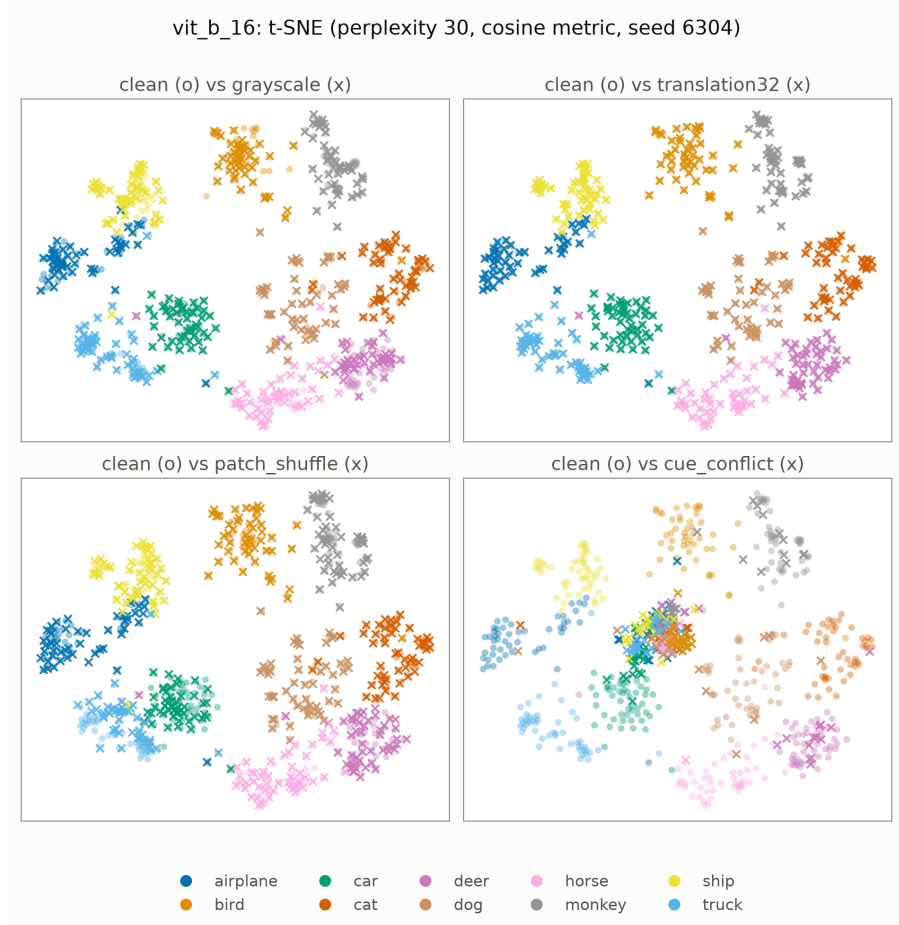


Figure 7: Vit-B t-SNE: Representations overlap except in Cue Conflict

D Task 2: Per-Method Results

D.1 Source Only ERM

Table 16: Source only ERM - the main domain gap identified

Domain	Accuracy (%)	Macro-F1 (%)
Photo (source, val)	97.30	96.85
Art Painting (source, val)	91.71	91.69
Cartoon (source, val)	95.74	96.00
Mean source	94.91	94.85
Worst source	91.71	91.69
Sketch (target)	61.59	63.09

Table 17: Target Domain by class - high amounts of variation.

Class	Sketch accuracy (%)
Dog	21.11
Elephant	94.73
Giraffe	77.95
Guitar	70.07
Horse	46.32
House	83.75
Person	61.25

D.2 DAN - MMD Alignment

Table 18: DA with MMD - Slight improvement over Source Only

	Source-only	DAN ($\lambda_{\text{MMD}} = 1$)
Photo val acc (%)	97.30	95.80
Art Painting val acc (%)	91.71	93.41
Cartoon val acc (%)	95.74	95.95
Mean source acc (%)	94.91	95.05
Mean source macro-F1 (%)	94.85	94.93
Sketch acc (%)	61.59	64.44
Sketch macro-F1 (%)	63.09	60.55
Δ Sketch acc	—	+2.85
Domain separability (%)	100.00	81.87

Table 19: MMD by Class - Good by class improvement

Class	Source-only (%)	DAN (%)	Δ
Dog	21.11	47.93	+26.82
Elephant	94.73	75.54	−19.19
Giraffe	77.95	61.62	−16.33
Guitar	70.07	74.18	+4.11
Horse	46.32	57.60	+11.28
House	83.75	100.00	+16.25
Person	61.25	86.25	+25.00

D.3 DANN - Adversarial Alignment

Table 20: DANN - The performances worsened

	Source-only	DANN
Photo val acc (%)	97.30	95.80
Art Painting val acc (%)	91.71	92.93
Cartoon val acc (%)	95.74	93.60
Mean source acc (%)	94.91	94.11
Mean source macro-F1 (%)	94.85	94.03
Sketch acc (%)	61.59	55.97
Sketch macro-F1 (%)	63.09	58.23
Δ Sketch acc	—	−5.62
Domain separability (%)	100.00	99.31

Table 21: Tracking Per Epoch - Loss was decreasing while Domain accuracy was going to 0.5

Epoch	Cls. loss	Domain loss	Domain acc.	Mean source F1 (%)
1	0.533	0.384	0.817	91.37
2	0.234	0.597	0.673	89.19
3	0.159	0.669	0.597	94.03
4	0.261	0.742	0.542	84.62
5	0.176	0.686	0.551	92.80
6	0.084	0.681	0.541	92.37
7	0.070	0.694	0.502	93.92
8	0.081	0.694	0.501	93.90

Selected epoch in bold. Chance domain accuracy is 0.5 and $\ln 2 = 0.693$.

D.4 CDAN - Class-Conditional Adversarial Alignment

Table 22: CDAN: Significant Improvement from before

	Source-only	CDAN
Photo val acc (%)	97.30	97.30
Art Painting val acc (%)	91.71	91.46
Cartoon val acc (%)	95.74	95.52
Mean source acc (%)	94.91	94.76
Mean source macro-F1 (%)	94.85	94.66
Sketch acc (%)	61.59	74.65
Sketch macro-F1 (%)	63.09	73.85
Δ Sketch acc	—	+13.06
Domain separability (%)	100.00	99.86

Table 23: Two most wrong - Similar pattern

True class	Source-only	DANN	CDAN
Dog	elephant 333, giraffe 167	person 397, house 18	person 139, elephant 62
Horse	giraffe 293, elephant 111	dog 204, person 166	dog 134, person 33
Elephant	person 21, giraffe 13	person 400, dog 49	person 65, dog 58
Giraffe	elephant 99, horse 34	person 164, dog 142	dog 175, horse 56

Two most frequent wrong predictions per class, with counts.

E Task 3: Additional Results

E.1 DAN-DG - Pairwise Source-Domain Alignment

Table 24: Comparison with ERM - complete model collapsed.

	ERM	DAN-DG ($\lambda_{DG} = 1$)
Photo val macro-F1 (%)	96.85	5.86
Art Painting val macro-F1 (%)	91.69	5.14
Cartoon val macro-F1 (%)	96.00	4.21
Mean source acc (%)	94.91	21.68
Mean source macro-F1 (%)	94.85	5.07
Worst source macro-F1 (%)	91.69	4.21
Sketch acc (%)	61.59	4.07
Sketch macro-F1 (%)	63.09	1.12
Δ Sketch acc	—	−57.52
Source-domain separability (%)	91.21	38.74
Δ_{sharp}	0.2095	0.0433

Table 25: λ Ramp - Improvement from before but still collapsing

Variant	λ_{DG}	Source (%)		Sketch (%)		Sep. (%)	Δ_{sharp}
		Mean F1	Worst F1	Acc.	Δ		
ERM (no alignment)	0	94.85	91.69	61.59	—	91.21	0.2095
As specified	1	5.07	4.21	4.07	−57.52	38.74	0.0433
Reduced weight	0.1	96.78	96.29	69.81	+8.22	72.80	0.1394
Increased weight	10	5.07	4.21	4.07	−57.52	27.47	0.0057
λ ramp	0 to 1	94.43	92.41	62.31	+0.72	76.37	0.3216
Bandwidth gradient	1	94.45	92.56	71.49	+9.90	70.05	0.2400

Table 26: Step Data - Model is destroyed by step 50.

Step	λ_{DG}	Cls. loss	MMD	$\ f\ $	Feature std
0	1	2.042	0.725	27.93	0.7891
50	1	1.948	0.286	3.43	0.0017
100	1	1.932	0.625	2.37	0.0003
150	1	1.925	0.525	2.10	0.0000
199	1	1.959	0.748	2.03	0.0000
199	0	0.096	—	61.48	1.6616

Identical code and seed; only λ_{DG} differs. $\ln 7 = 1.946$ is the cross-entropy of a uniform predictor over seven classes. Gradient clipping at 1.0 did not prevent it.

E.2 SAM - Parameter-Space Stability

Table 27: SAM - Stable performances and great improvement

	ERM	SAM ($\rho = 0.05$)
Photo val macro-F1 (%)	96.85	96.41
Art Painting val macro-F1 (%)	91.69	93.65
Cartoon val macro-F1 (%)	96.00	97.13
Mean source acc (%)	94.91	95.74
Mean source macro-F1 (%)	94.85	95.73
Worst source macro-F1 (%)	91.69	93.65
Sketch acc (%)	61.59	74.12
Sketch macro-F1 (%)	63.09	74.98
Δ Sketch acc	—	+12.53
Source-domain separability (%)	91.21	89.01
Δ_{sharp}	0.2095	0.1116

F Task 4: Additional Results

F.1 Vanilla Closed-Set Baseline

Table 28: Vanilla Model - Intentionally made strong with 95 validation accuracy for later comparison.

Vanilla ResNet-18	
Epochs	100
Selected epoch	93
CIFAR-10 train accuracy (%)	100.00
CIFAR-10 validation accuracy (%)	95.08
CIFAR-10 test accuracy, CSA (%)	94.26

CIFAR-10 known classes; 45,000 / 5,000 stratified train / validation split, seed 6304. Checkpoint selected by validation accuracy only.

F.2 Post-Hoc Novelty Score

Table 29: TODO: your caption and key take-away.

Score	τ	Test acc. (%)	Near rej. (%)	Far rej. (%)	FPR@95TPR near / far (%)
MSP	0.1721	94.34	28.25	45.38	71.75 / 54.62
MLS	-5.7175	94.77	30.62	54.62	69.38 / 45.38
Energy	-5.9177	94.84	31.38	54.88	68.62 / 45.12
Mahalanobis	2764.0347	94.62	27.25	50.12	72.75 / 49.88

τ is the 95th percentile of unknownness on the CIFAR-10 validation set; accept when $u(x) \leq \tau$. Rejection is the share of unknowns not accepted; FPR@95TPR is the share incorrectly accepted.

F.3 GCSC - Strong Closed-Set Classifier + MLS

Table 30: GCSC - Validations acc and Test acc both increase slightly

	Vanilla	GCSC
Selected epoch	93	98
CIFAR-10 validation accuracy (%)	95.08	95.62
CIFAR-10 test accuracy, CSA (%)	94.26	95.38

Identical recipe, initialization, optimiser, schedule, batch size, epochs and seed; GCSC adds RandAugment(2, 9) after the crop and flip.

Table 31: GCSC - Classifications scores all improve

Model	AUROC near (%)	AUROC far (%)	AUROC all (%)	CSA (%)
Vanilla + MLS	79.42	89.84	84.63	94.26
GCSC + MLS	81.84	92.13	86.99	95.38
Δ	+2.42	+2.29	+2.36	+1.12

F.4 PROSER - Classifier and Data Placeholders

Table 32: PROSER - Same Level of CIFAR-10 for less EPOCH

	Vanilla	GCSC	PROSER
Epochs	100	100	50
Selected epoch	93	98	14
CIFAR-10 validation accuracy (%)	95.08	95.62	94.72
CIFAR-10 test accuracy, CSA (%)	94.26	95.38	93.63

PROSER is initialised from the selected Vanilla checkpoint with five dummy classifiers appended. CSA uses the ten known-class logits only.

Table 33: PROSER - Same Level of performance for less EPOCH.

Model	Score	AUROC near (%)	AUROC far (%)	AUROC all (%)	CSA (%)
Vanilla	MLS	79.42	89.84	84.63	94.26
GCSC	MLS	81.84	92.13	86.99	95.38
PROSER	MLS	77.61	87.57	82.59	93.63
PROSER	Placeholder	74.36	84.82	79.59	93.63